

Lecture #15

Content to be covered

- ❑ Polarimeter
 - ❖ Half-Shade Device
 - ❖ Laurent's Half Shade Polarimeter
 - ❖ Bi-Quartz Polarimeter

Resources to be consulted

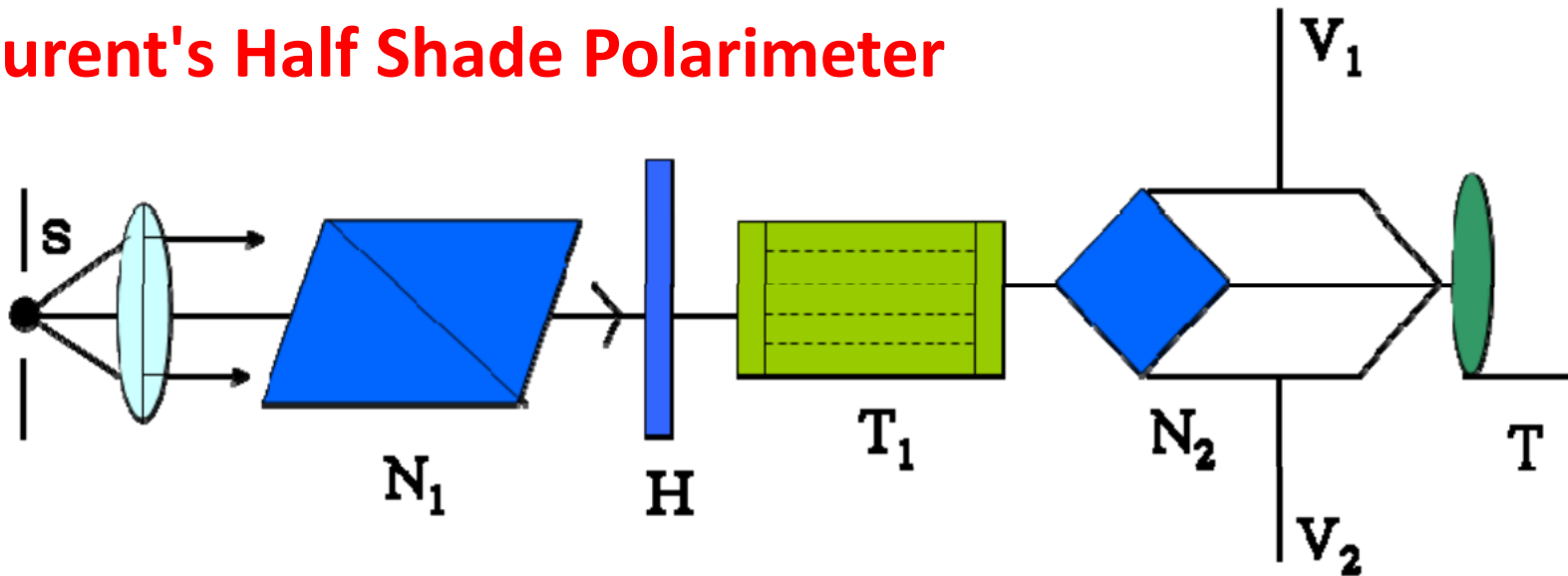
- ❑ OPTICS by Ajay Ghatak

Polarimeter

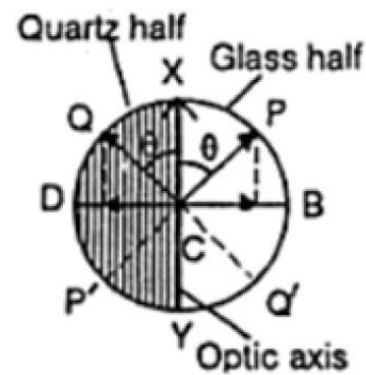
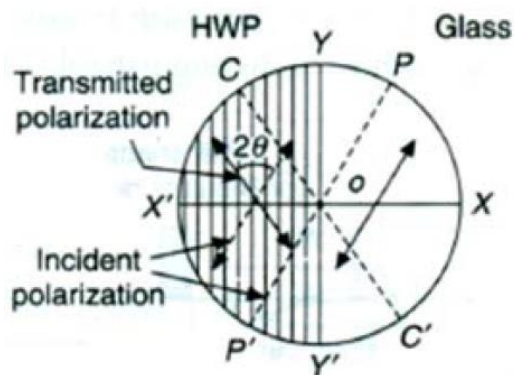
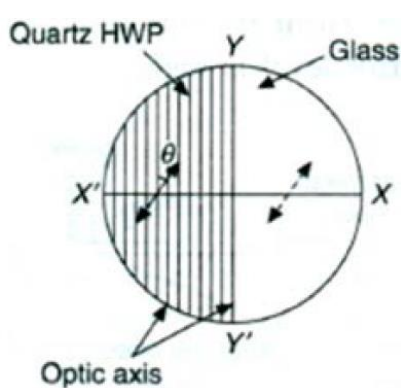
A device designed for accurate measurement of angle of rotation of plane of vibration of a plane polarized light by an optically active medium is said to be a polarimeter.

1. Laurent's Half shade polarimeter
2. Bi-quartz polarimeter

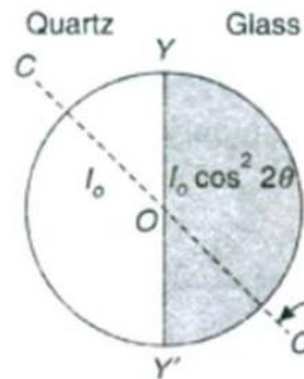
Laurent's Half Shade Polarimeter



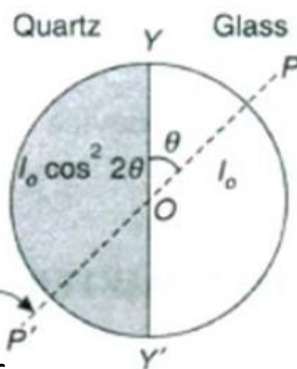
Half-Shade Device (H)



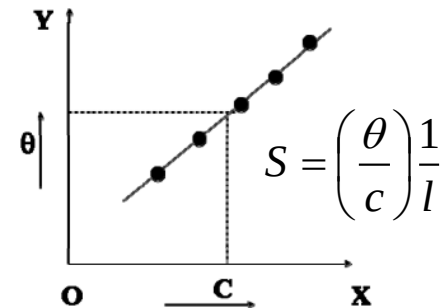
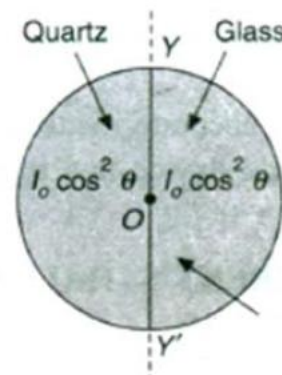
It consists of a circular plate with one half made up of quartz (**HWP**) cut parallel to the optic axis which is parallel to YY' . Other half is made up of glass, so that glass absorbs same light as quartz plate does.



pass axis of analyzer

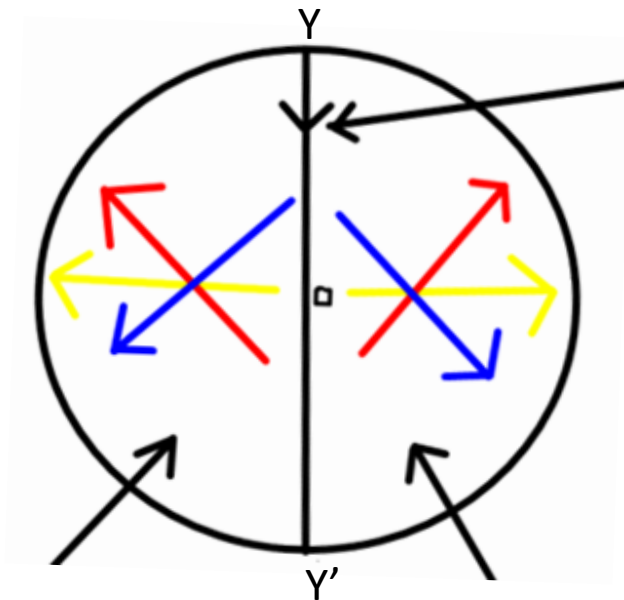
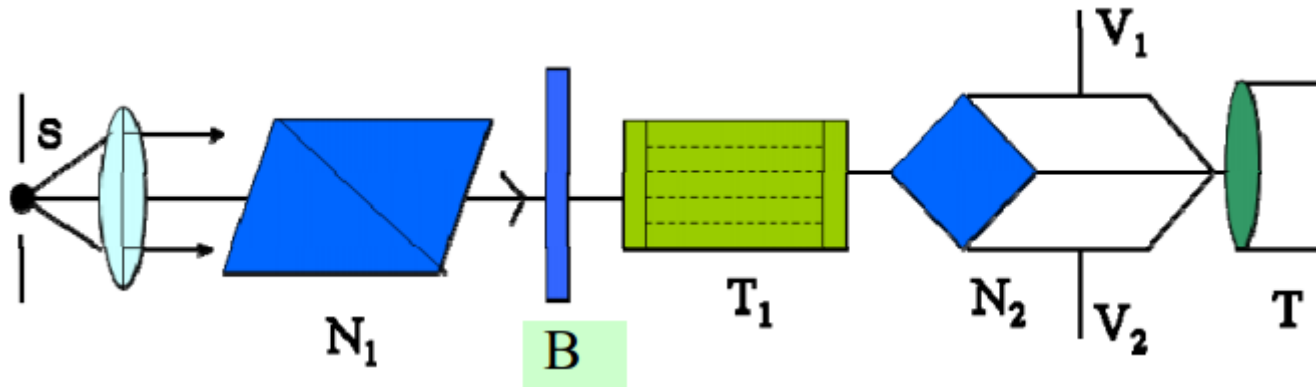


pass axis of analyzer



where l is length of tube T_1 in dm.

Bi-Quartz Polarimeter

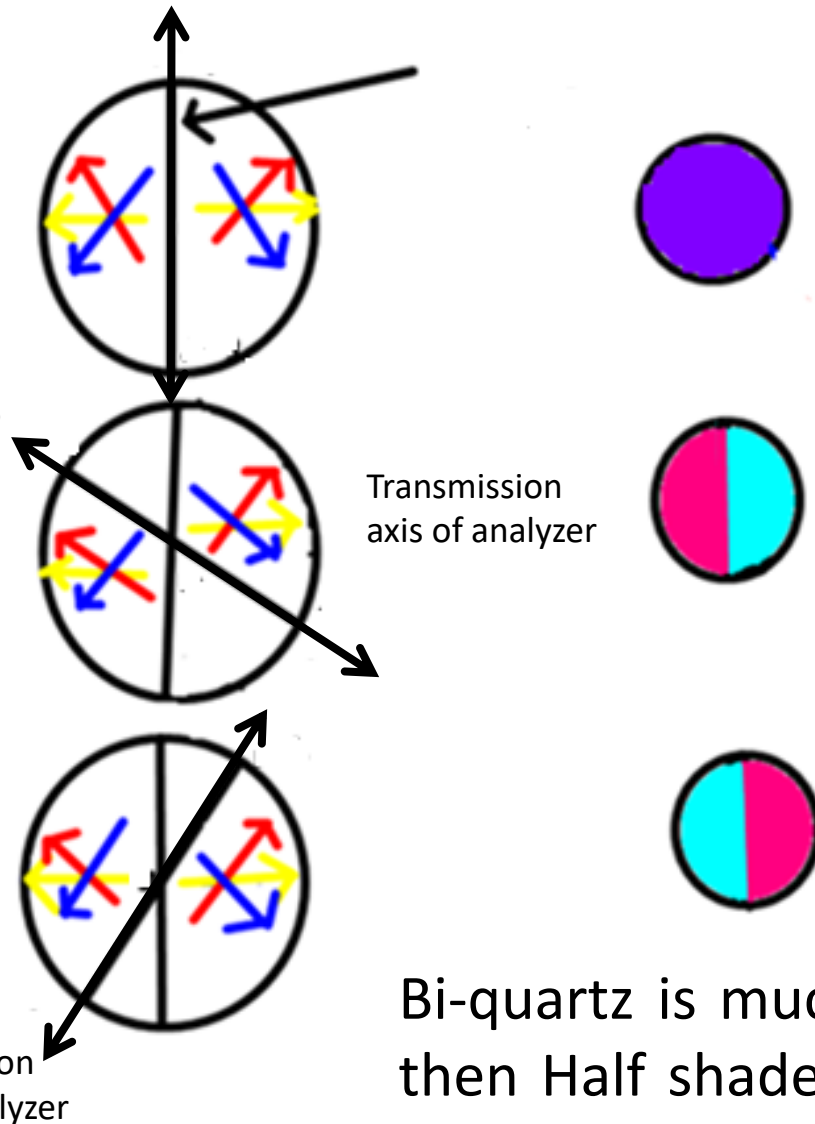


Left handed Quartz

Right handed Quartz

$$\theta = \frac{\pi}{\lambda} (\mu_L \sim \mu_R) t$$

$$S = \frac{\theta}{lc}$$



- White light source is used.
- Two semicircular quartz plates (Right and Left handed) with Optic axis perpendicular to crystal surface (rotation effect only).
- HWP for yellow light.
- Incident PPL vibrates always along the YY'. (YY' is tint of passes.)
- If Analyzer axes is $\perp$ to YY' then yellow color will be disappeared and we can get resultant of Red and Blue colour (Reddish violet color) as min intensity.
- If Analyzer pass axis is $\parallel$ to YY' then same colour of equal intensity will be observed on both side.

Bi-quartz is much more sensitive and accurate than Half shade polarimeter. But having major drawback for color blind person.